

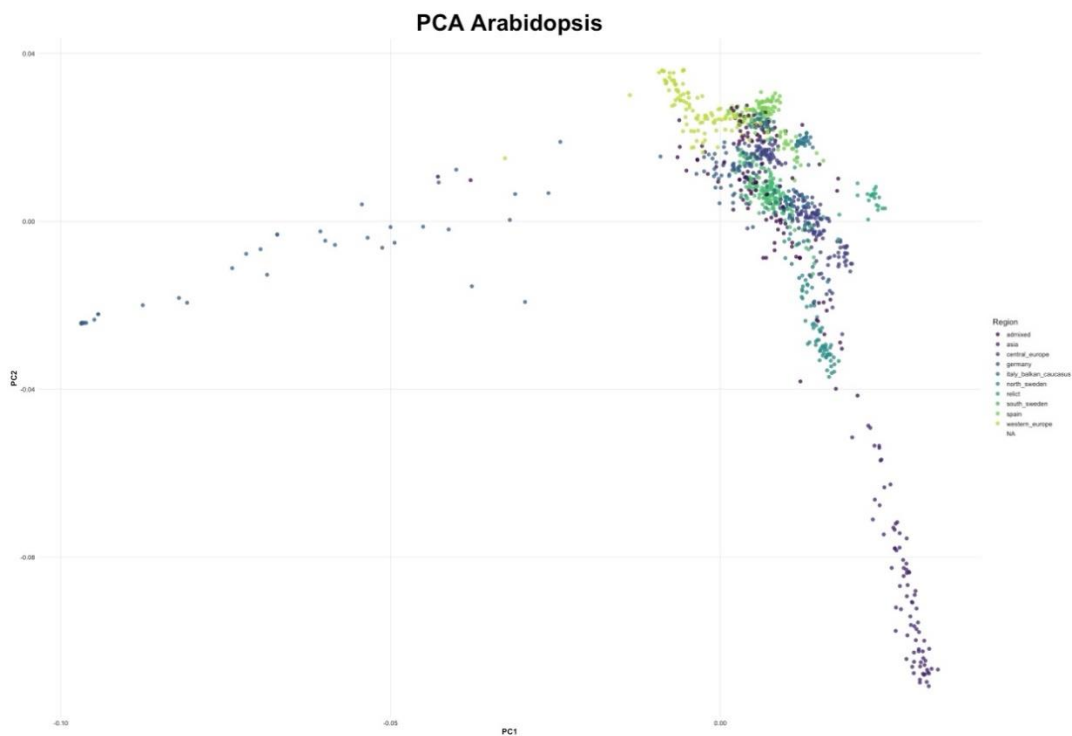
PCA Assignment

Questions-

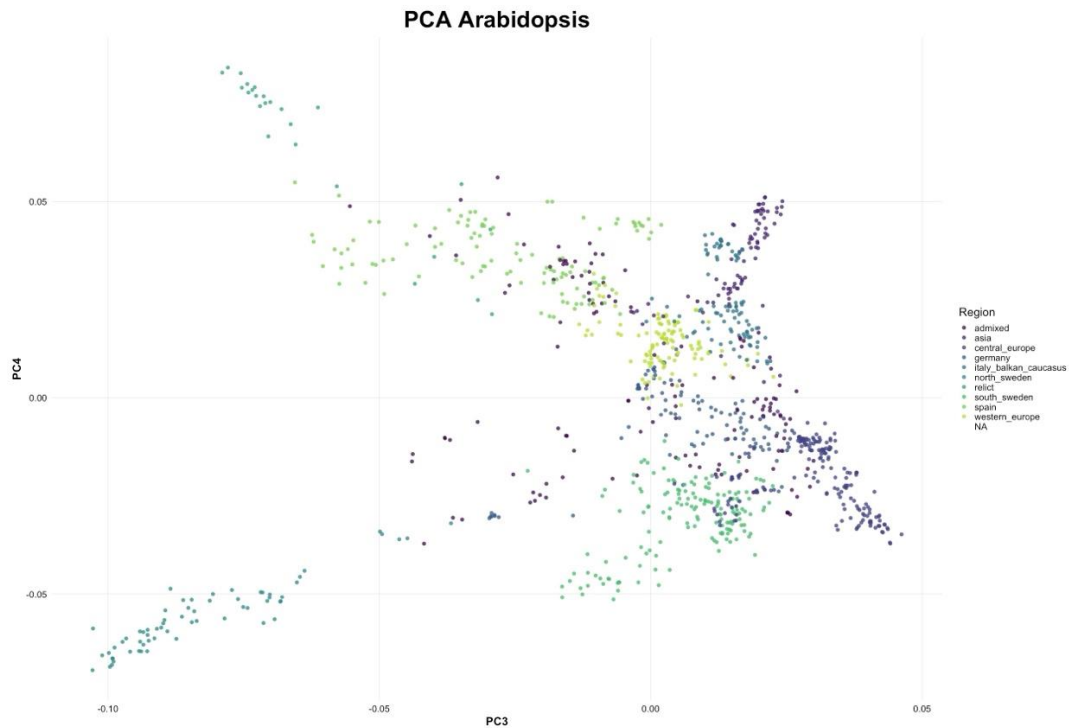
1. Based on the plot you've made, what are the major axes of variation in your diversity dataset?
How do you know this?

In the PCA plot of Arabidopsis, the first principal component (PC1) shows the biggest difference among the samples. This means most of the genetic variation is along PC1. The second component (PC2) shows some extra differences, but not as much as PC1. I can tell this because the samples are spread the most along the PC1 axis, forming clear groups based on their regions.

PCA 1



PCA 2



2. How many sites were in the dataset of all SNPs that you used to calculate the PCA without subsampling? How did you find this out?

The dataset used to compute the PCA contained 82,877,803 SNP sites. This information was obtained directly from the SNP summary output (**.eigenvec** and associated files generated during the PCA process). The total site count was provided in the summary output produced by PLINK before the PCA computation.

3. How did changing the number of SNPs affect the PCA plots you made? How many SNPs would you recommend that another researcher use for Arabidopsis PCA?

When fewer SNPs were used, the PCA plot showed fewer clear patterns, hence the groups of samples were harder to see. When more SNPs were included, the plots became clearer and showed stronger separation between different regions or populations depicting clear genetic differences. For Arabidopsis, it's good to use around 50,000–100,000 good-quality SNPs for PCA. That number gives enough detail to see population structure patterns without making the analysis too heavy for the computer.